LAPLACE TRANSFORMS

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DEFINITION

The Laplace transform of a function f(t), for all $t \geq 0$ is defined by the integral

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}dt = F(s)$$

The domain of F(s) is the set of all values of s for which this integral converges. The Laplace transform of f is denoted by both F and $\mathcal{L}$.

Examples

Example 1: Let $f(t) = 1$. The Laplace transform satisfies:

$$\begin{aligned}\mathcal{L}[1] &= \int_0^\infty 1e^{-st}dt \\ &= -\frac{1}{s}e^{-st}\Big|_0^\infty \\ &= -\frac{1}{s}[e^{-\infty} - e^0] \\ &= -\frac{1}{s}[-1] \\ &= \frac{1}{s}\end{aligned}$$

Example 2: Let $f(t) = e^{at}$. The Laplace transform satisfies:

$\mathcal{L}[e^{at}] = \int_0^\infty e^{at}e^{-st}dt$

$$= \int_0^{\infty} e^{-(s-a)t} dt$$
$$= -\frac{1}{(s-a)} e^{-(s-a)t} \Big|_0^{\infty}$$
$$-\frac{1}{(s-a)} [e^{-\infty} - e^0]$$
$$= \frac{1}{(s-a)}$$

Example 3: Let $f(t) = sin(at); t \leq 0$. But
$sin(at) = \frac{1}{2i}(e^{iat} - e^{-iat})$
$\mathcal{L}[sin(at)] = \frac{1}{2i} \int_0^{\infty} (e^{iat} - e^{-iat}) e^{-st} dt$
$$= \frac{1}{2i} \int_0^{\infty} e^{iat} e^{-st} dt - \frac{1}{2i} \int_0^{\infty} e^{-iat} e^{-st} dt$$
$$= \frac{1}{2i} [\int_0^{\infty} e^{-(s-ia)t} dt - \int_0^{\infty} e^{-(s+ia)t} dt]$$
$$= \frac{1}{2i} [\frac{1}{s-ia} - \frac{1}{s+ia}]$$

$$= \frac{a}{a^2+s^2}$$

Example 4: Let $f(t) = cos(at)$. But

$cos(at) = \frac{1}{2}(e^{iat} + e^{-iat})$

$$\mathcal{L}[cos(at)] = \frac{1}{2}\int_0^\infty (e^{iat} + e^{-iat})e^{-st}dt$$

$$= \frac{1}{2}\int_0^\infty e^{iat}e^{-st}dt + \frac{1}{2i}\int_0^\infty e^{-iat}e^{-st}dt$$

$$= \frac{1}{2}[\int_0^\infty e^{-(s-ia)t}dt + \int_0^\infty e^{-(s+ia)t}dt]$$

$$= \frac{1}{2}[\frac{1}{s-ia} + \frac{1}{s+ia}]$$

$$= \frac{s}{a^2+s^2}$$

The table below shows the Laplace transform of some functions:

f(t)	$F(s) = \mathcal{L}[f(t)]$
n,n=1,2,...	$\frac{n}{s}, s > 0$
e^{at}	$\frac{1}{s-a}, s > 0$
t^n	$\frac{n!}{s^{n+1}}, s > 0$
sin(at)	$\frac{a}{s^2+a^2}, s > 0$
cos(at)	$\frac{s}{a^2+s^2}, s > 0$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}} s > 0$
$e^{bt} sin(at)$	$\frac{a}{(s-b)^2+a^2}$
$e^{bt} cos(at)$	$\frac{s-b}{(s-b)^2+a^2}$
sinh(at)	$\frac{a}{s^2-a^2}$
cosh(at)	$\frac{s}{s^2-a^2}$

PROPERTIES

LINEARITY

One property of Laplace transform is its linearity.
$\mathcal{L}(af + bg) = a\mathcal{L}(f) + b\mathcal{L}(g)$:

Property 2

If $\mathcal{L}f(t) = F(s)$, then for any constant **a**
$\mathcal{L}[e^{at}f(t)] = F(s - a)$

Property 3

if $\mathcal{L}[f(t)] = F(s)$, then for any positive integer **n**,
$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}[F(s)]$

Find the Laplace of the given functions.

EXAMPLES

1. $g(t) = e^{-2t}cos3t$
2. $p(t) = sin(4t)$
3. $h(t) = e^{3t} + cos(6t) + e^{-3t}cos(6t)$

SOLUTION

1. $g(t) = \mathcal{L}[e^{-2t}cos3t]$
$= \frac{s-(-2)}{(s-(-2))^2+3^2}$
$= \frac{s+2}{(s+2)^2+9}$

2. $g(t) = \mathcal{L}[sin(4t)]$
$= \frac{4}{s^2+16}$

3. $g(t) = \mathcal{L}[e^{3t} + cos(6t) + e^{-3t}cos(6t)]$
$= \mathcal{L}[e^{3t}] + \mathcal{L}[cos(6t)] + \mathcal{L}[e^{-3t}cos(6t)]$
$= \frac{3}{s-3} + \frac{s}{s^2+36} + \frac{s+3}{(s+3)^2+36}$

Trial

Solve the following Laplace transforms

1. $\mathcal{L}\{e^{-7t}t^2\}$
2. $\mathcal{L}\ \{te^{-4t}\}$
3. $\mathcal{L}\{e^{-2t}sin5t\}$
4. $\mathcal{L}\{tcos(\sqrt{7}t)\}$
5. $\mathcal{L}\{tsinh2t\}$
6. $\mathcal{L}\{-2e^{-2t}sin4t + 4e^{-2t}cos4t\}$
7. $\mathcal{L}\{tcos2t\}$

PIECEWISE CONTINUOUS FUNCTION

Definition (Piecewise function)

A function f is said to be piecewise continuous on an interval $\alpha \leq t \leq \beta$ if the interval can be partitioned by a finite number of points $\alpha = t_0 < t_1 < ... < t_n = \beta$ so that:

1. $\mathcal{F}$ is continuous on each sub-interval $t_{i-1} < t < t_i$, and
2. f approaches a finite limit as the endpoints of each sub-interval are approached from within the sub-interval.

Example

Define the piecewise continuous function:

$$g(t) = \begin{cases} e^{2t} & 0 \leq t \leq 1 \\ 4 & 1 < t \end{cases}$$

The Laplace transform satisfies:

$$\begin{aligned} &= \int_0^1 e^{-st} e^{2t} dt + \int_1^\infty 4e^{-st} dt \\ &= \int_0^1 e^{-(s-2)t} dt + \int_1^\infty 4e^{-st} dt \\ &= -\frac{1}{s-2} e^{-(s-2)t} \Big|_0^1 - \frac{4}{s} e^{-st} \Big|_1^\infty \end{aligned}$$

$\frac{1-e^{-(s-2)}}{s-2} + \frac{4e^{-s}}{s}, s > 0, s \neq 2$

Trial

Solve the following piecewise continuous functions using Laplace transforms.

- $$f(t) = \begin{cases} t^2 & 0 \leq t \leq 1 \\ t+2 & 1 < t \leq 2 \\ 6-t & 2 < t \leq 3 \end{cases}$$
- $$h(t) = \begin{cases} t & 0 \leq t \leq 1 \\ 3-t & 1 < t \leq 2 \\ 1 & 2 < t \leq 3 \end{cases}$$

Trial

- $$g(t) = \begin{cases} t^2 & 0 \leq t \leq 1 \\ 1 & 1 < t \leq 2 \\ 3 - t & 2 < t \leq 3 \end{cases}$$

INVERSE LAPLACE

DEFINITION

An inverse Laplace transform of F(s), designated by $\mathcal{L}^{-1}\{F(s)\}$, is another function f(t) having the property that $\mathcal{L}\{f(t)\} = F(s)$.

PROPERTY (LINEARITY)

If the inverse Laplace transforms of two functions $F(s)$ and $G(s)$ exist, then for any constants c_1 and c_2,

$$\mathcal{L}^{-1}\{c_1F(s) + c_2G(s)\} = c_1\mathcal{L}^{-1}\{F(s)\} + c_2\mathcal{L}^{-1}\{G(s)\}$$

EXAMPLES

- $\mathcal{L}^{-1}\{\frac{1}{s}\}$
  Here $F(s) = \frac{1}{s}$. From our previous studies, we have
  $\mathcal{L}\{1\} = 1/s$.
  Therefore,
  $\mathcal{L}^{-1}\{1/s\} = 1$

- $\mathcal{L}^{-1}\{\frac{1}{s-8}\}$
  Here $F(s) = \frac{1}{s-8}$. From our previous studies, we have
  $\mathcal{L}\{e^{8t}\} = \frac{1}{s-8}$.
  Therefore,
  $\mathcal{L}^{-1}\{\frac{1}{s-8}\} = e^{8t}$

- $\mathcal{L}^{-1}\{\frac{s+1}{s^2-9}\}$
$= \mathcal{L}^{-1}\{\frac{s}{s^2-9}\} + \mathcal{L}^{-1}\{\frac{1}{s^2-9}\}$
$= cosh(3t) + \mathcal{L}^{-1}\{\frac{1}{3}(\frac{3}{s^2-9})\}$
$= cosh(3t) + \frac{1}{3}sinh(3t)$

- $\mathcal{L}^{-1}\{\frac{6}{s} - \frac{1}{s-8} + \frac{4}{s-3}\}$
$= 6 - e^{8t} + 4e^{3t}$

TRIAL

Solve the following inverse Laplace transforms:

- $\mathcal{L}^{-1}\{\frac{8}{3s^2+12} + \frac{3}{s^2-49}\}$
- $\mathcal{L}^{-1}\{\frac{6s}{s^2+25} + \frac{3}{s^2+25}\}$
- $\mathcal{L}^{-1}\{\frac{s}{(s-2)^2+9}\}$ Hint:(Use create and destroy)

THE LAPLACE TRANSFORM OF DERIVATIVES

Theorem 1

Suppose that f is continuous and f^{-1} is piecewise continuous on any interval $0 \leq t \leq A$. Suppose further that there exist constants K, a, and M such that $|f(t)| \leq Ke^{at}$ for $t \geq M$. Then $\mathcal{L}[f'(t)]$ exists for $s > a$, and moreover

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$$

THEOREM 2

Suppose that the functions $f, f', ..., f^{(n-1)}$ are continuous and that $f^{(n)}$ is piecewise continuous on any interval $0 \leq t \leq A$. Suppose further that there exist constants K, a, and M such that $|f(t)| \leq Ke^{at}$, $|f'(t)| \leq Ke^{at}$,..., $|f^{n-1}(t)| \leq Ke^{at}$ for t $\geq M$. Then $\mathcal{L}\{f^{(n)}(t)\}$ exists for $s > a$ and is given by:

$$\mathcal{L}\{f^n(t)\} = s^n\mathcal{L}\{f(t)\} - s^{n-1}f(0) - ... - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

Examples

Solve the following differential equations using Laplace transforms

1. $y' - 5y = e^{5t}, y(0) = 0$

SOLUTION

$\mathcal{L}\{y'\} - 5\mathcal{L}\{y\} = 0$

$sy(s) - y(0) - 5y(s) = \frac{1}{s-5}$

$(s-5)y(s) - 0 = \frac{1}{s-5}$

$y(s) = \frac{1}{(s-5)(s-5)}$

$y(s) = \frac{1}{(s-5)^2}$

Using the idea of Laplace inverse;

$y(s) = te^{5t}$

2. Solve $y'' + 4y = 0, y(0) = 2, y'(0) = 2$

SOLUTION

$s^2y(s) - sy(0) - y'(0) + 4y(s) = 0$

$(s^2 + 4)y(s) - 2s - 2 = 0$

$y(s) = \frac{2s+2}{s^2+4}$

$y(s) = \frac{2s}{s^2+4} + \frac{2}{s^2+4}$

Applying Laplace inverse;

$y(t) = 2cos(2t) + sin(2t)$

TRIAL

Solve the following using Laplace transforms;

1. $y'' - y' - 2y = 4t^2, y(0) = 1, y'(0) = 4$
2. $y''' + y' = e^t, y(0) = y'(0) = y''(0) = 0$
3. $\frac{dI}{dt} + 5I = 5, I(0) = 0$